Final topologies of Lipschitz Yoneda-continuous maps

Abstract

Let (X, d) be a quasi-metric space, and let $\mathcal{L}_\alpha(X, d)$ be the space of α-Lipschitz Yoneda-continuous maps from X to the extended non-negative reals $\overline{\mathbb{R}}_+$. Write $\mathcal{L}_\infty(X, d) = \bigcup_{\alpha>0} \mathcal{L}_\alpha(X, d)$, with the topology inherited from the Scott topology on the dcpo of lower semicontinuous maps $X \to \overline{\mathbb{R}}_+$. We study when this topology is the final topology for the inclusions of the fixed Lipschitz layers.

The principal result is that the final-topology property holds for every continuous Yoneda-complete quasi-metric space. The algebraic case is proved by finite cone interpolation and a compactification of a finitely branching tree of cone functions; the general case follows from the representation of continuous Yoneda-complete spaces as 1-Lipschitz Yoneda-continuous retracts of algebraic ones. For ordinary metric spaces, the property is shown to be equivalent to metric completeness.

For an arbitrary quasi-metric space we record exact categorical equivalent conditions, notably that the natural map from the coproduct of the fixed layers is a quotient map. For a continuous, not necessarily complete, quasi-metric space we give a more structural criterion: the final-topology property is equivalent to continuity of the simultaneous extension map into the formal-ball completion. Finally, we show that Yoneda-completeness and Lipschitz regularity are not necessary in the unrestricted quasi-metric category.

Keywords: quasi-metric space, Yoneda completeness, formal ball, Scott topology, final topology, Lipschitz map
2020 MSC: 54E99, 54C35, 06B35, 54D30

1. Introduction and statement of the problem

Throughout, $\mathbb{R}_+ = [0, +\infty)$ denotes the finite non-negative reals and $\overline{\mathbb{R}}_+ = [0, +\infty]$ the extended non-negative reals. A quasi-metric is a map $d : X \times X \to \overline{\mathbb{R}}_+$ such that

$$d(x, x) = 0, \qquad d(x, z) \le d(x, y) + d(y, z),$$

and $d(x, y) = d(y, x) = 0$ implies $x = y$. Thus we adopt the usual T_0 separation convention. On $\overline{\mathbb{R}}_+$ we use

$$d_{\mathbb{R}}(u, v) = \begin{cases} 0, & u \le v, \\ u - v, & v < u < +\infty, \\ +\infty, & u = +\infty > v. \end{cases}$$

A formal ball has a *finite* radius:

$$\mathbf{B}(X, d) = X \times \mathbb{R}_+, \qquad (x, r) \le^{d^+} (y, s) \iff d(x, y) \le r - s. \tag{1}$$

The d-Scott topology on X is the topology induced by $x \mapsto (x, 0)$ from the Scott topology on $\mathbf{B}(X, d)$. A net $(x_i)_{i\in I}$ is Cauchy when, for every $\varepsilon > 0$, there is i_0 such that $d(x_i, x_j) < \varepsilon$ whenever $i_0 \le i \le j$. A point x is its d-limit when

$$d(x, y) = \limsup_i d(x_i, y) \qquad (y \in X),$$

where $\limsup_i u_i = \inf_i \sup_{j\ge i} u_j$ for a net in $\overline{\mathbb{R}}_+$. The space is *Yoneda-complete* when every Cauchy net has a d-limit. An α-Lipschitz map, where always $0 < \alpha < +\infty$ in this paper, satisfies

$$d_{\mathbb{R}}(f(x), f(y)) \le \alpha d(x, y),$$

with the convention $\alpha \cdot (+\infty) = +\infty$. It is Yoneda-continuous when, whenever a Cauchy net (x_i) has d-limit x, the image net $(f(x_i))$ has $d_{\mathbb{R}}$-limit $f(x)$ (the image net is Cauchy because f is Lipschitz). On standard quasi-metric spaces this is equivalent to Scott continuity of

$$\mathbf{B}_\alpha(f) : \mathbf{B}(X,d) \longrightarrow \mathbf{B}(\overline{\mathbb{R}}_+, d_{\mathbb{R}}), \qquad (x,r) \longmapsto (f(x), \alpha r),$$

or, equivalently for $\overline{\mathbb{R}}_+$-valued maps, of $(x,r) \mapsto f(x) - \alpha r$; see [4, Lemmas 6.3 and 6.4] and [2, Lemma 2.7].

Write $\mathcal{L}(X)$ for the dcpo of maps $X \to \overline{\mathbb{R}}_+$ that are continuous from the d-Scott topology to the Scott topology of $\overline{\mathbb{R}}_+$ (equivalently, lower semicontinuous), ordered pointwise and equipped with its Scott topology. For finite $\alpha > 0$, define

$$\mathcal{L}_\alpha(X,d) = \{f \in \mathcal{L}(X) : f \text{ is } \alpha\text{-Lipschitz and Yoneda-continuous}\},$$

with its subspace topology, and put

$$\mathcal{L}_\infty(X,d) = \bigcup_{0<\alpha<+\infty} \mathcal{L}_\alpha(X,d).$$

Thus $\mathcal{L}_\infty(X,d)$ also carries the subspace topology inherited from $\mathcal{L}(X)$. On a standard quasi-metric space, the formal-ball criterion implies that an α-Lipschitz d-Scott-continuous map is automatically Yoneda-continuous, but the displayed definition is used for arbitrary quasi-metric spaces. The question whether the topology of $\mathcal{L}_\infty(X,d)$ is determined by its fixed Lipschitz layers was posed by Goubault-Larrecq and is still displayed as an open problem at the time of writing [8].

We say that (X,d) has the *final-topology property* when the topology of $\mathcal{L}_\infty(X,d)$ is determined by the family $(\mathcal{L}_\alpha(X,d))_{\alpha>0}$, namely when

$$[\, U \cap \mathcal{L}_\alpha(X,d) \text{ is open in } \mathcal{L}_\alpha(X,d) \text{ for every } \alpha > 0 \,] \quad \Longrightarrow \quad U \text{ is open in } \mathcal{L}_\infty(X,d). \tag{2}$$

The reverse implication is automatic. Equivalently, $\mathcal{L}_\infty(X,d)$ is the colimit of the directed system formed by the inclusions $\mathcal{L}_\alpha(X,d) \hookrightarrow \mathcal{L}_\beta(X,d)$, $\alpha \le \beta$, in the category of topological spaces. The following formal characterizations are valid without completeness or continuity assumptions.

Proposition 1.1. *For every quasi-metric space (X,d), the following conditions are equivalent.*

(i) *(X,d) has the final-topology property.*

(ii) *The natural continuous surjection*

$$q_X : \coprod_{n\ge1} \mathcal{L}_n(X,d) \longrightarrow \mathcal{L}_\infty(X,d), \qquad q_X(n,f) = f,$$

is a quotient map.

(iii) *For every topological space Z and every map $F : \mathcal{L}_\infty(X,d) \to Z$, F is continuous if and only if $F|_{\mathcal{L}_n(X,d)}$ is continuous for every $n \ge 1$.*

(iv) *The implication in* (iii) *holds with Z restricted to the Sierpiński space $\mathbb{S}$.*

(v) *A subset $C \subseteq \mathcal{L}_\infty(X,d)$ is closed if and only if $C \cap \mathcal{L}_n(X,d)$ is closed in $\mathcal{L}_n(X,d)$ for every $n \ge 1$.*

Proof. The inclusions $\mathcal{L}_\alpha(X,d) \subseteq \mathcal{L}_\beta(X,d)$ for $0 < \alpha \le \beta$ show that the positive integers are cofinal. The final topology for the maps $\mathcal{L}_n(X,d) \to \mathcal{L}_\infty(X,d)$ is precisely the quotient topology induced by q_X, which proves the equivalence of (i) and (ii). The universal property of a final topology gives (iii). Open subsets are classified by continuous maps to $\mathbb{S}$, giving (iv), and taking complements gives (v). □

The next elementary observation will be used twice.

Lemma 1.2. *Let $X_\infty = \bigcup_{n\ge1} X_n$ and $Y_\infty = \bigcup_{n\ge1} Y_n$ be increasing unions of subspaces. Suppose there are continuous maps*

$$e : X_\infty \longrightarrow Y_\infty, \qquad r : Y_\infty \longrightarrow X_\infty, \qquad r \circ e = \mathrm{id}_{X_\infty},$$

such that $e(X_n) \subseteq Y_n$ and $r(Y_n) \subseteq X_n$ for every n. If the topology of Y_∞ is final with respect to the Y_n, then the topology of X_∞ is final with respect to the X_n.

Proof. Let $U \subseteq X_\infty$ have open intersection with every X_n. Then $r^{-1}(U) \cap Y_n = (r|_{Y_n})^{-1}(U \cap X_n)$ is open in Y_n for every n, hence $r^{-1}(U)$ is open in Y_∞. Since $U = e^{-1}(r^{-1}(U))$, the set U is open. □

The main result is the following.

Theorem 1.3. *Every continuous Yoneda-complete quasi-metric space has the final-topology property.*

In particular, the answer to the metric-space version is affirmative for every complete metric space. In fact, completeness is exact in that case.

Theorem 1.4. *Let (X, d) be an ordinary metric space. Then (X, d) has the final-topology property if and only if it is complete.*

Theorem 1.3 is the substantive answer to the original open problem. The metric theorem identifies the precise necessity statement in the symmetric setting. Later, theorem 6.3 gives a completion-theoretic necessary and sufficient condition for continuous quasi-metric spaces that need not be complete.

The order in (1) is the specialization order of the formal-ball quasi-metric

$$d^+((x, r), (y, s)) = \max\{d(x, y) - r + s, 0\}.$$

A quasi-metric space is *standard* when, for every directed family $((x_i, r_i))_{i\in I}$ of formal balls and every $t \in \mathbb{R}_+$, the family $((x_i, r_i))_{i\in I}$ has a supremum if and only if the shifted family $((x_i, r_i + t))_{i\in I}$ has a supremum. In that case, if $\sup_i(x_i, r_i) = (x, r)$, then $r = \inf_i r_i$, and $\sup_i(x_i, r_i + t) = (x, r + t)$ [4, Definition 2.1 and Proposition 2.4]. Every metric space, every Yoneda-complete quasi-metric space, and every order quasi-metric is standard [4, Proposition 2.2]. A standard quasi-metric space is *continuous* when $\mathbf{B}(X, d)$ is a continuous poset [4, Definition 3.10].

A point $x \in X$ is a *center point* when every open d^+-ball $B^{d^+}_{(x,0),<\varepsilon}$ is Scott-open in $\mathbf{B}(X, d)$. The space has *enough center points* when those balls, with x a center point and $\varepsilon > 0$, form a base of the Scott topology on $\mathbf{B}(X, d)$ [4, Definitions 5.5 and 5.11]. For standard spaces, having enough center points is equivalent to algebraicity [4, Theorem 5.16]; this is the characterization of "algebraic" used below.

The proof of theorem 1.3 uses the following established facts.

(F1) If $0 < \alpha \leq \beta$, then $\mathcal{L}_\alpha(X, d) \subseteq \mathcal{L}_\beta(X, d)$. This follows directly from the Lipschitz inequality, while lower semicontinuity and Yoneda-continuity do not depend on the chosen constant; compare [4, Proposition 6.7(5)]. Hence the positive integers are cofinal and $\mathcal{L}_\infty(X, d) = \bigcup_{n\geq 1} \mathcal{L}_n(X, d)$.

(F2) If (X, d) is continuous Yoneda-complete, the topology on every $\mathcal{L}_\alpha(X, d)$ is the topology of pointwise convergence, while the Scott topology on $\mathcal{L}(X)$ is the compact-open topology [2, Propositions 7.7 and 8.1]; see also [3].

(F3) A standard algebraic quasi-metric space has enough center points [4, Theorem 5.16].

(F4) If (X, d) is standard, x is a center point, and $a < \infty$, then

$$y \longmapsto \max\{a - d(x, y), 0\}$$

is 1-Lipschitz Yoneda-continuous. Finite maxima of such functions are the least 1-Lipschitz maps satisfying the corresponding lower bounds at the centers [2, Lemmas 6.19 and 6.20]. Scaling and finite maxima preserve Lipschitz Yoneda-continuity [2, Proposition 2.8(1),(3)].

(F5) Every continuous Yoneda-complete quasi-metric space is a 1-Lipschitz Yoneda-continuous retract of an algebraic Yoneda-complete quasi-metric space [4, Theorem 7.9].

The main construction is a compact space of paths through a finitely branching tree of cone functions, obtained from a set that is open on every fixed Lipschitz layer.

2. Preliminaries on cones and final-open sets

On a standard quasi-metric space, for a positive integer n, a center point x, and a finite number $a > 0$, define the n-cone by the subtraction-free formula

$$\kappa_{n,x,a}(y) = \min\{t \in \overline{\mathbb{R}}_+ : a \le t + nd(x,y)\}. \tag{3}$$

Equivalently, it is $\max\{a - nd(x,y), 0\}$ when $d(x,y) < +\infty$, and it is 0 when $d(x,y) = +\infty$. By (F4), applied to a/n and then multiplied by n, $\kappa_{n,x,a} \in \mathcal{L}_n(X,d)$. It is the least n-Lipschitz map g such that $g(x) \ge a$.

We first record the order-theoretic consequence of layerwise openness.

Lemma 2.1. *Let $U \subseteq \mathcal{L}_\infty(X,d)$ satisfy the hypothesis on the left-hand side of (2). Then U is upward closed for the pointwise order.*

Proof. Let $g \in U$ and $g \le h$. By cofinality, choose an integer n such that $g, h \in \mathcal{L}_n(X,d)$. The specialization order on $\mathcal{L}_n(X,d)$ is the pointwise order. Indeed, $\mathcal{L}_n(X,d)$ is a subspace of the Scott space $\mathcal{L}(X)$, whose specialization order is the pointwise order, and specialization order restricts to every subspace. Equivalently, if $u \le v$ pointwise, every ambient Scott-open neighborhood of u contains v; conversely, if $u \not\le v$ pointwise, some point-evaluation open $\{w : w(x) > r\}$ contains u but not v. Every open subset is upward closed for the specialization order. Since $U \cap \mathcal{L}_n(X,d)$ is open and contains g, it also contains h. □

The next lemma is the finite approximation mechanism used at every level of the construction.

Lemma 2.2. *Let (X,d) be a standard algebraic quasi-metric space, let $n \ge 1$, and let $q \in \mathcal{L}_n(X,d)$. Suppose that $m \ge 1$, $y_1, \dots, y_m \in X$, and $0 \le b_i < q(y_i)$ for every i. Then there are center points $x_1, \dots, x_m$ and finite numbers $a_1, \dots, a_m > 0$ such that*

$$h = \max_{1 \le i \le m} \kappa_{n,x_i,a_i}$$

satisfies

$$h \le q, \qquad h(y_i) > b_i \quad (1 \le i \le m), \qquad a_i < q(x_i) \quad (1 \le i \le m).$$

Proof. For $q \in \mathcal{L}_n(X,d)$, the map (see [2, Lemma 2.7])

$$q^\sharp : \mathbf{B}(X,d) \longrightarrow \mathbb{R} \cup \{+\infty\}, \qquad q^\sharp(z,r) = q(z) - nr,$$

is Scott-continuous. For each i, the Scott-open set

$$W_i = (q^\sharp)^{-1}(\,]b_i, +\infty]\,)$$

contains $(y_i, 0)$. Since X has enough center points, there are a center point x_i and $\varepsilon_i > 0$ such that

$$(y_i, 0) \in B^{d^+}_{(x_i,0),<\varepsilon_i} \subseteq W_i.$$

Choose t_i with

$$d(x_i, y_i) < t_i < \varepsilon_i.$$

Then $(x_i, t_i) \in W_i$, and hence

$$q(x_i) - nt_i > b_i.$$

We may therefore choose a finite a_i satisfying

$$b_i + nd(x_i, y_i) < a_i < q(x_i).$$

It follows that $\kappa_{n,x_i,a_i}(y_i) > b_i$. Since q is n-Lipschitz and $q(x_i) > a_i$, the minimality of the cone gives $\kappa_{n,x_i,a_i} \le q$. Taking the finite maximum proves the claim. □

Lemma 2.3. *Let $(T_n)_{n \ge n_0}$ be nonempty finite sets and let $\pi_n : T_{n+1} \to T_n$ be parent maps. The set $\widehat{T}$ of all nonempty finite compatible paths and all infinite compatible paths is compact Hausdorff when finite paths are isolated and neighborhoods of an infinite path are its finite-prefix cylinders.*

Proof. For each $n \geq n_0$, adjoin a terminal symbol † to T_n and consider the compact product $\prod_{n\geq n_0}(T_n \cup \{\dagger\})$, with discrete finite factors. The desired space is identified with the subset of sequences whose first coordinate is not †, whose nonterminal coordinates satisfy the parent relations, and which remain terminal after the first occurrence of †. Every one of these conditions is closed, so the subset is compact Hausdorff. A finite path is isolated by prescribing its last node and the following terminal coordinate, while the product cylinders at nonterminating sequences are exactly the finite-prefix cylinders. □

3. The algebraic Yoneda-complete case

Theorem 3.1. *Every algebraic Yoneda-complete quasi-metric space has the final-topology property.*

Proof. An algebraic Yoneda-complete quasi-metric space is standard [4, Proposition 2.2] and continuous [4, Proposition 5.18]. Hence the pointwise-topology statement in (F2) and theorem 2.2 both apply. Let $U \subseteq \mathcal{L}_\infty(X,d)$ be open on every layer, and let $f \in U$. By theorem 2.1, U is upward closed. If $\mathbf{0} \in U$, then $U = \mathcal{L}_\infty(X,d)$, so assume from now on that $\mathbf{0} \notin U$.

Choose an integer $n_0 \geq 1$ such that $f \in \mathcal{L}_{n_0}(X,d)$. Since the topology on $\mathcal{L}_{n_0}(X,d)$ is pointwise, there are $m \geq 1$, points $y_1, \dots, y_m$, and thresholds $0 \leq b_i < f(y_i)$ such that

$$B_0 = \{g \in \mathcal{L}_{n_0}(X,d) : g(y_i) > b_i \ (1 \leq i \leq m)\} \subseteq U \cap \mathcal{L}_{n_0}(X,d).$$

The constraint list is nonempty because $\mathbf{0} \notin U$. Applying theorem 2.2, we obtain a nonzero function

$$h_{n_0} = \max_{s \in T_{n_0}} \kappa_{n_0,x_s,a_s} \tag{4}$$

with T_{n_0} finite and nonempty, such that $h_{n_0} \in B_0$ and

$$h_{n_0} \leq f, \qquad h_{n_0} \in U \cap \mathcal{L}_{n_0}(X,d), \qquad 0 < a_s < f(x_s) \quad (s \in T_{n_0}).$$

Suppose recursively that

$$h_n = \max_{s \in T_n} \kappa_{n,x_s,a_s} \in U \cap \mathcal{L}_n(X,d) \tag{5}$$

has been constructed, where T_n is finite and nonempty. Regard h_n as an element of $\mathcal{L}_{n+1}(X,d)$. Since $\mathbf{0} \notin U$, choose $m_n \geq 1$, points $y_{n,j}$, and thresholds $0 \leq b_{n,j} < h_n(y_{n,j})$ such that

$$B_n = \{g \in \mathcal{L}_{n+1}(X,d) : g(y_{n,j}) > b_{n,j} \ (1 \leq j \leq m_n)\} \subseteq U \cap \mathcal{L}_{n+1}(X,d).$$

Apply theorem 2.2 with $q = h_n$. This gives

$$h_{n+1} = \max_{t \in T_{n+1}} \kappa_{n+1,x_t,a_t} \in B_n \subseteq U \cap \mathcal{L}_{n+1}(X,d) \tag{6}$$

with

$$h_{n+1} \leq h_n, \qquad a_t < h_n(x_t) \text{ for every } t \in T_{n+1}. \tag{7}$$

Thus

$$f \geq h_{n_0} \geq h_{n_0+1} \geq \cdots.$$

For every $t \in T_{n+1}$, choose a parent $\pi(t) = s \in T_n$ for which

$$a_t < a_s - nd(x_s, x_t).$$

Such an s exists by (7) and the representation (5); the right-hand side must be positive. Hence

$$a_t + nd(x_s, x_t) < a_s. \tag{8}$$

The finite sets T_n, together with the parent maps, form a finitely branching rooted forest with finitely many roots.

Let $\widehat{T}$ be the compact Hausdorff path space supplied by theorem 2.3.

Along an infinite path

$$\beta = (s_{n_0}, s_{n_0+1}, \ldots),$$

put

$$p_n = \left(x_{s_n}, \frac{a_{s_n}}{n}\right) \in \mathbf{B}(X, d).$$

By (8),

$$d(x_{s_n}, x_{s_{n+1}}) < \frac{a_{s_n} - a_{s_{n+1}}}{n} < \frac{a_{s_n}}{n} - \frac{a_{s_{n+1}}}{n+1},$$

so $p_n \leq^{d^+} p_{n+1}$. If

$$A = \max_{s \in T_{n_0}} a_s,$$

then every descendant height is at most A, and consequently the radii of p_n tend to zero. Since the space is continuous and Yoneda-complete, $\mathbf{B}(X, d)$ is a continuous dcpo [4, Theorem 3.7]; hence the chain has a supremum $\sup_n p_n = (x_\beta, r_\beta)$. Standardness then gives $r_\beta = \inf_n a_{s_n}/n = 0$. Hence

$$\sup_n p_n = (x_\beta, 0) \tag{9}$$

for a uniquely determined point $x_\beta \in X$.

Define

$$\phi : \widehat{T} \longrightarrow X$$

by $\phi(s) = x_s$ on a finite path ending in s, and $\phi(\beta) = x_\beta$ on an infinite path. The map ϕ is continuous. Indeed, continuity at finite paths is immediate. Let O be a d-Scott open neighborhood of x_β. Choose a Scott-open subset $\mathcal{O} \subseteq \mathbf{B}(X, d)$ with $O = \mathcal{O} \cap X$. Since the supremum in (9) belongs to $\mathcal{O}$, some p_N belongs to $\mathcal{O}$. Every finite or infinite path extending s_N is mapped to a point z such that $p_N \leq^{d^+} (z, 0)$: for a finite descendant this follows from $p_N \leq^{d^+} \cdots \leq^{d^+} p_m \leq^{d^+} (x_{s_m}, 0)$, and for an infinite descendant it follows from the supremum formula (9). Upward closure of $\mathcal{O}$ therefore shows that the entire cylinder at s_N is mapped into O.

Define a height function $a : \widehat{T} \to [0, A]$ by assigning a_s to the finite path ending in s, and

$$a(\beta) = \lim_{n \to \infty} a_{s_n}$$

to an infinite path. The limit exists because heights decrease along every branch. The function a is upper semicontinuous: if $a(\beta) < c$, then some node on β has height $< c$, and all of its descendants have still smaller height; finite paths are isolated.

Because the root set is finite and $a_s < f(x_s)$ at every root, choose $\delta > 0$ such that

$$a_s + \delta \leq f(x_s) \qquad (s \in T_{n_0}). \tag{10}$$

We claim that

$$f(\phi(\tau)) \geq a(\tau) + \delta \qquad (\tau \in \widehat{T}). \tag{11}$$

For a finite descendant t, we have $a_t < h_{n_0}(x_t)$, so one of the root cones, say the one indexed by s, attains the positive value $h_{n_0}(x_t)$ at x_t. In particular, $d(x_s, x_t) < +\infty$. Using the n_0-Lipschitz inequality for f and (10),

$$\begin{aligned} f(x_t) &\geq f(x_s) - n_0 d(x_s, x_t) \\ &\geq a_s + \delta - n_0 d(x_s, x_t) \\ &= h_{n_0}(x_t) + \delta > a_t + \delta. \end{aligned}$$

The same assertion for roots is exactly (10).

For an infinite path $\beta = (s_n)$, the map

$$(x, r) \longmapsto f(x) - n_0 r$$

is Scott-continuous on $\mathbf{B}(X,d)$. Applying it to (9) gives

$$\begin{aligned} f(x_\beta) &= \sup_n \left(f(x_{s_n}) - \frac{n_0 a_{s_n}}{n} \right) \\ &\geq \sup_n \left(a_{s_n} + \delta - \frac{n_0 a_{s_n}}{n} \right) \\ &\geq \lim_n \left(a_{s_n} + \delta - \frac{n_0 a_{s_n}}{n} \right) = a(\beta) + \delta. \end{aligned}$$

This proves (11).

Choose $0 < \varepsilon < \delta$, and put

$$M = \left\lfloor \frac{A}{\varepsilon} \right\rfloor.$$

For $0 \leq k \leq M$, let

$$C_k = \{\tau \in \widehat{T} : a(\tau) \geq k\varepsilon\}, \qquad K_k = \phi(C_k), \qquad Q_k = \mathrm{sat}(K_k).$$

Upper semicontinuity of a makes C_k closed in the compact space $\widehat{T}$, hence K_k is compact in X. The saturation of a compact set is compact in every topological space: a finite subfamily covering K_k has open union containing K_k, and therefore contains $\mathrm{sat}(K_k)$. Thus Q_k is compact saturated. Moreover, the open set $f^{-1}(]k+1)\varepsilon, +\infty])$ contains K_k by (11), hence it also contains Q_k. Consequently

$$N = \bigcap_{k=0}^{M} \{g \in \mathcal{L}_\infty(X,d) : g(x) > (k+1)\varepsilon \text{ for every } x \in Q_k\} \tag{12}$$

is the intersection with $\mathcal{L}_\infty(X,d)$ of a compact-open, equivalently Scott-open, subset of $\mathcal{L}(X)$. It is therefore a neighborhood of f in the inherited topology on $\mathcal{L}_\infty(X,d)$.

We finally show that $N \subseteq U$. Let $g \in N$, and choose an integer $n \geq n_0$ with $g \in \mathcal{L}_n(X,d)$. For $s \in T_n$, set

$$k = \left\lfloor \frac{a_s}{\varepsilon} \right\rfloor.$$

Then the finite path ending in s belongs to C_k, so $x_s \in K_k \subseteq Q_k$ and

$$g(x_s) > (k+1)\varepsilon > a_s.$$

By the minimality of the cone, $\kappa_{n,x_s,a_s} \leq g$ for every $s \in T_n$. Hence $h_n \leq g$. Since $h_n \in U$ and U is upward closed, $g \in U$. Thus $f \in N \subseteq U$, and U is open. □

4. Passage to continuous Yoneda-complete spaces

The final-topology property descends along the retracts occurring in (F5).

Lemma 4.1. *Let (X,d) and (Y,∂) be standard quasi-metric spaces, and suppose that (X,d) is a 1-Lipschitz Yoneda-continuous retract of (Y,∂): there are 1-Lipschitz Yoneda-continuous maps*

$$e : X \longrightarrow Y, \qquad r : Y \longrightarrow X, \qquad r \circ e = \mathrm{id}_X .$$

If (Y,∂) has the final-topology property, then so does (X,d).

Proof. Precomposition gives maps

$$\begin{aligned} E : \mathcal{L}_\infty(X,d) &\longrightarrow \mathcal{L}_\infty(Y,\partial), \quad E(f) = f \circ r, \\ R : \mathcal{L}_\infty(Y,\partial) &\longrightarrow \mathcal{L}_\infty(X,d), \quad R(g) = g \circ e. \end{aligned}$$

They preserve every integer Lipschitz layer and satisfy $R \circ E = \mathrm{id}_{\mathcal{L}_\infty(X,d)}$. Since the spaces are standard, the formal-ball criterion shows that e and r are continuous for the respective d-Scott topologies. Hence precomposition sends lower semicontinuous maps to lower semicontinuous maps. On the ambient function dcpos it is monotone and preserves directed pointwise suprema, so it is Scott-continuous; its restrictions to $\mathcal{L}_\infty(X,d)$ and to every layer are therefore continuous for the inherited topologies. Apply theorem 1.2. □

Proof of theorem 1.3. By (F5), (X, d) is a 1-Lipschitz Yoneda-continuous retract of an algebraic Yoneda-complete quasi-metric space. Apply theorems 3.1 and 4.1. □

Corollary 4.2. *Let (X, d) be continuous Yoneda-complete and let Z be any topological space. A map $F : \mathcal{L}_\infty(X, d) \to Z$ is continuous if and only if every restriction $F|_{\mathcal{L}_\alpha(X,d)}$ is continuous.*

Proof. This is the universal property of the final topology established in theorem 1.3. □

5. Completeness is necessary and sufficient for metric spaces

We now prove theorem 1.4. The positive direction follows from theorem 1.3, because every metric space is continuous and metric completeness agrees with Yoneda-completeness. The converse is independent of the preceding construction.

Let (X, d) be an incomplete metric space, let $(\widehat{X}, \widehat{d})$ be its completion, and choose $\xi \in \widehat{X} \setminus X$. Every finite-valued Lipschitz map $f : X \to [0, \infty)$ has a unique Lipschitz extension $\widehat{f}$ to $\widehat{X}$. A Lipschitz map to $\overline{\mathbb{R}}_+$ which takes the value $+\infty$ at one point is identically $+\infty$, so define

$$E_\xi(f) = \begin{cases} \widehat{f}(\xi), & f \not\equiv +\infty, \\ +\infty, & f \equiv +\infty. \end{cases}$$

Lemma 5.1. *For every $\alpha > 0$, the restriction $E_\xi|_{\mathcal{L}_\alpha(X,d)} : \mathcal{L}_\alpha(X, d) \to \overline{\mathbb{R}}_+$ is continuous for the pointwise topology, and hence for the inherited subspace topology, which contains all point-evaluation opens.*

Proof. Fix $r \geq 0$ and $f \in \mathcal{L}_\alpha(X, d)$ with $E_\xi(f) > r$. First suppose that f is finite-valued and write $L = E_\xi(f)$. Choose $x \in X$ so close to ξ that

$$2\alpha\widehat{d}(x, \xi) < L - r.$$

Then

$$f(x) \geq L - \alpha\widehat{d}(x, \xi) > r + \alpha\widehat{d}(x, \xi).$$

The pointwise neighborhood

$$\{g \in \mathcal{L}_\alpha(X, d) : g(x) > r + \alpha\widehat{d}(x, \xi)\}$$

is contained in $E_\xi^{-1}(\,]r, +\infty]\,)$, since

$$E_\xi(g) \geq g(x) - \alpha\widehat{d}(x, \xi) > r.$$

If $f \equiv +\infty$, choose any $x \in X$ and any finite $T > r + \alpha\widehat{d}(x, \xi)$. The point-evaluation neighborhood $\{g : g(x) > T\}$ contains f and is mapped into $]r, +\infty]$ by the same estimate. □

Fix $0 < r < 1$ and set

$$U_r = E_\xi^{-1}(\,]r, +\infty]\,) \subseteq \mathcal{L}_\infty(X, d).$$

By theorem 5.1, U_r is open on every layer. We show that it is not open in $\mathcal{L}_\infty(X, d)$.

Let $\mathcal{F}$ be the directed set of nonempty finite subsets of X, ordered by inclusion. For $F \in \mathcal{F}$, define

$$\delta_F = \min_{x \in F} \widehat{d}(x, \xi) > 0, \qquad g_F(x) = \min\left\{1, \frac{\widehat{d}(x, \xi)}{\delta_F}\right\}.$$

Then g_F is $1/\delta_F$-Lipschitz, hence belongs to $\mathcal{L}_\infty(X, d)$, and its extension satisfies $E_\xi(g_F) = 0$. If $F \subseteq G$, then $\delta_G \leq \delta_F$ and $g_F \leq g_G$. Moreover,

$$\sup_{F \in \mathcal{F}} g_F = \mathbf{1}$$

pointwise, because $g_{\{x\}}(x) = 1$ for every $x \in X$.

If U_r were open in the subspace topology inherited from the Scott topology on $\mathcal{L}(X)$, there would be a Scott-open $O \subseteq \mathcal{L}(X)$ with $U_r = O \cap \mathcal{L}_\infty(X, d)$. The constant function $\mathbf{1}$ belongs to O, and it is the directed supremum of the family $(g_F)_{F \in \mathcal{F}}$. Scott openness would force some g_F to belong to O, contradicting $E_\xi(g_F) = 0$. Thus U_r is layerwise open but not open in $\mathcal{L}_\infty(X, d)$, so the final-topology property fails. This completes the proof of theorem 1.4.

6. Extension to the formal-ball completion

We now give a non-tautological necessary and sufficient condition for a continuous quasi-metric space that is not assumed Yoneda-complete. This criterion explains exactly how the points added by completion may create new global topology on the union of the Lipschitz layers.

Let (X, d) be continuous. Write

$$S = S(X, d)$$

for its formal-ball completion. Here $(x, r) \prec (y, s)$ means $d(x, y) < r - s$, a rounded ideal is a nonempty $\prec$-directed, $\prec$-downward-closed subset of $\mathbf{B}(X, d)$, and its aperture is the infimum of its (finite) radii. The points of $S(X, d)$ are the aperture-zero rounded ideals, equipped with the formal-ball-completion quasi-metric. This space is algebraic Yoneda-complete. The canonical isometric embedding is

$$\eta(x) = \{b \in \mathbf{B}(X, d) : b \prec (x, 0)\}.$$

The universal property of S states that every α-Lipschitz map from X into a Yoneda-complete quasi-metric space has a unique α-Lipschitz Yoneda-continuous extension to S [4, Section 7]; see also [5, 6].

For $f \in \mathcal{L}_\alpha(X, d)$, write $\widehat{f} : S \to \overline{\mathbb{R}}_+$ for this extension. The following formula and the second, noncanonical map from X to S are central.

Lemma 6.1. *Let (X, d) be continuous, $\alpha > 0$, and $f \in \mathcal{L}_\alpha(X, d)$. For every $D \in S(X, d)$,*

$$\widehat{f}(D) = \sup_{(x,r)\in D} (f(x) - \alpha r). \tag{13}$$

Here the supremum is taken in $\mathbb{R} \cup \{+\infty\}$; because D has aperture zero, its value lies in $\overline{\mathbb{R}}_+$.

Define

$$s : X \longrightarrow S(X, d), \qquad s(x) = \{b \in \mathbf{B}(X, d) : b \ll (x, 0)\}. \tag{14}$$

Then s is 1-Lipschitz Yoneda-continuous and

$$\widehat{f}(s(x)) = f(x) \qquad (x \in X). \tag{15}$$

Proof. Since X is continuous, it is standard. By the formal-ball criterion for Yoneda-continuity, the map

$$f^\sharp : \mathbf{B}(X, d) \longrightarrow \mathbb{R} \cup \{+\infty\}, \qquad f^\sharp(x, r) = f(x) - \alpha r,$$

is Scott-continuous.

We spell out the completion calculation. There is an order isomorphism

$$\sigma_X : \mathbf{B}(S(X, d)) \longrightarrow \mathrm{RI}(\mathbf{B}(X, d), \prec), \qquad \sigma_X(D, t) = D + t,$$

where $D + t = \{(x, r + t) : (x, r) \in D\}$ [4, Section 7]. Under this isomorphism, $\mathbf{B}_\alpha(\widehat{f})$ is the unique Scott-continuous extension of $\mathbf{B}_\alpha(f)$ from the abstract basis $\mathbf{B}(X, d)$ to its rounded-ideal completion. Every rounded ideal D is the directed supremum of the principal rounded ideals generated by its elements, so the value of that extension at D is the directed supremum of the values $\mathbf{B}_\alpha(f)(x, r)$ for $(x, r) \in D$. Hence, at an aperture-zero ideal D,

$$(\widehat{f}(D), 0) = \sup_{(x,r)\in D} (f(x), \alpha r) \quad \text{in } \mathbf{B}(\overline{\mathbb{R}}_+, d_{\mathbb{R}}). \tag{16}$$

To read this supremum, use the order isomorphism

$$\Theta : \mathbf{B}(\overline{\mathbb{R}}_+, d_{\mathbb{R}}) \longrightarrow \{(u, v) \in (\mathbb{R} \cup \{+\infty\}) \times (-\infty, 0] : u - v \geq 0\}, \qquad \Theta(z, t) = (z - t, -t).$$

The displayed target has the inherited product order and is closed under directed suprema in that product. Hence its directed suprema are computed coordinatewise. The second coordinate of the supremum in (16) is

$$-\alpha \inf_{(x,r)\in D} r = 0,$$

while its first coordinate is

$$\sup_{(x,r)\in D} (f(x) - \alpha r).$$

This proves (13); it also shows directly that the displayed supremum is in $\overline{\mathbb{R}}_+$.

The map s is the map denoted η^0_S in [4, Lemma 7.5]. The poset of formal balls is continuous and $(x, 0)$ is the directed supremum of the formal balls way-below it. Scott continuity of $f^\sharp$ and the formula just proved yield

$$\widehat{f}(s(x)) = \sup_{(y,r)\ll(x,0)} f^\sharp(y, r) = f^\sharp(x, 0) = f(x).$$

□

If a function belongs to two Lipschitz layers, its two extensions agree: the extension with the smaller Lipschitz constant is admissible for the larger one, and uniqueness applies. Hence there is a well-defined map

$$\mathrm{Ext}_X : \mathcal{L}_\infty(X, d) \longrightarrow \mathcal{L}_\infty(S(X, d)), \qquad f \longmapsto \widehat{f}. \tag{17}$$

Lemma 6.2. *For every $\alpha > 0$, the restriction*

$$\mathrm{Ext}_{X,\alpha} : \mathcal{L}_\alpha(X, d) \longrightarrow \mathcal{L}_\alpha(S(X, d))$$

is continuous. Moreover,

$$R_X : \mathcal{L}_\infty(S(X, d)) \longrightarrow \mathcal{L}_\infty(X, d), \qquad R_X(h) = h \circ s,$$

is continuous, preserves every Lipschitz layer, and satisfies

$$R_X \circ \mathrm{Ext}_X = \mathrm{id}_{\mathcal{L}_\infty(X,d)}. \tag{18}$$

Proof. The formal-ball completion is algebraic Yoneda-complete, so the topology on $\mathcal{L}_\alpha(S(X, d))$ is the topology of pointwise convergence. For $D \in S(X, d)$ and $a \geq 0$, (13) gives

$$\begin{aligned}\mathrm{Ext}^{-1}_{X,\alpha}\{h : h(D) > a\} &= \left\{ f : \sup_{(x,r)\in D} (f(x) - \alpha r) > a \right\} \\ &= \bigcup_{(x,r)\in D} \{f : f(x) > a + \alpha r\}.\end{aligned}$$

Each set in the last union is open in the topology inherited from $\mathcal{L}(X)$, because point evaluation is Scott-continuous. Thus $\mathrm{Ext}_{X,\alpha}$ is continuous.

The map s is 1-Lipschitz Yoneda-continuous between standard spaces, so it is continuous for the corresponding d-Scott topologies and precomposition by s preserves all Lipschitz layers. On the ambient function dcpos, precomposition is monotone and preserves directed pointwise suprema, hence is Scott-continuous. Equation (18) is (15). □

We can now state the completion criterion.

Theorem 6.3. *Let (X, d) be a continuous quasi-metric space, not necessarily Yoneda-complete. Then the following are equivalent.*

(i) *(X, d) has the final-topology property.*

(ii) *The simultaneous extension map*

$$\mathrm{Ext}_X : \mathcal{L}_\infty(X, d) \longrightarrow \mathcal{L}_\infty(S(X, d))$$

is continuous.

Proof. Assume first that (X, d) has the final-topology property. By theorem 6.2, the restriction of Ext_X to every fixed layer is continuous. The universal property in theorem 1.1 therefore makes Ext_X continuous.

Conversely, assume that Ext_X is continuous, and let

$$A_\infty = \mathrm{Ext}_X(\mathcal{L}_\infty(X, d)) \subseteq \mathcal{L}_\infty(S(X, d)).$$

By (18), Ext_X is a homeomorphism from $\mathcal{L}_\infty(X, d)$ onto A_∞, with inverse $R_X|_{A_\infty}$. Put

$$A_n = A_\infty \cap \mathcal{L}_n(S(X, d)).$$

Then

$$A_n = \mathrm{Ext}_X(\mathcal{L}_n(X, d)). \tag{19}$$

Indeed, one inclusion is immediate. For the other, if $h = \mathrm{Ext}_X(f) \in A_n$, then $f = R_X(h) \in \mathcal{L}_n(X, d)$.

The map

$$P = \mathrm{Ext}_X \circ R_X : \mathcal{L}_\infty(S(X, d)) \longrightarrow A_\infty$$

is a continuous retraction and maps each $\mathcal{L}_n(S(X, d))$ into A_n. Since $S(X, d)$ is algebraic Yoneda-complete, theorem 3.1 says that its function space has the final topology with respect to its fixed layers. By theorem 1.2, the same is true of A_∞ with respect to the A_n. Transporting this property through the homeomorphism $\mathrm{Ext}_X : \mathcal{L}_\infty(X, d) \to A_\infty$ and using (19) proves the final-topology property for X. □

The criterion has an equivalent compact-open form. It is more geometric than the quotient-map characterization, although it is still expressed in terms of the completion.

Corollary 6.4. *Let (X, d) be continuous. It has the final-topology property if and only if, for every compact saturated subset $K \subseteq S(X, d)$ and every $a \geq 0$, the set*

$$\left\{ f \in \mathcal{L}_\infty(X, d) : \widehat{f}(D) > a \text{ for every } D \in K \right\} \tag{20}$$

is open in $\mathcal{L}_\infty(X, d)$.

Proof. The space $S(X, d)$ is continuous Yoneda-complete. Hence the Scott topology on its dcpo of lower semicontinuous $\overline{\mathbb{R}}_+$-valued maps is the compact-open topology, generated by the subbasic conditions $[K > a]$ with K compact saturated [2, Proposition 7.7]. The set in (20) is exactly $\mathrm{Ext}_X^{-1}([K > a])$. Apply theorem 6.3. □

Corollary 6.5. *For an ordinary metric space (X, d), the simultaneous extension map to the Cauchy completion is continuous if and only if X is complete.*

Proof. Every metric space is continuous, and its formal-ball completion is naturally isomorphic to its usual Cauchy completion [4, Section 7]. Combine theorems 1.4 and 6.3. □

7. Necessary conditions in the quasi-metric setting

The metric characterization should not be extrapolated verbatim to all quasi-metric spaces.

Proposition 7.1. *Continuous Yoneda-completeness is not a necessary condition for the final-topology property in the full quasi-metric category.*

Proof. Let $P = \mathbb{N}$ with its usual order, and equip it with the order quasi-metric

$$d_\leq(m, n) = \begin{cases} 0, & m \leq n, \\ +\infty, & m \not\leq n. \end{cases}$$

The net $(n)_{n \in \mathbb{N}}$ is Cauchy, since $d_\leq(n, m) = 0$ whenever $n \leq m$. It has no $d_\leq$-limit: if $x \in \mathbb{N}$ were such a limit, then at $y = x$ the defining equation would read

$$0 = d_\leq(x, x) = \limsup_n d_\leq(n, x) = +\infty,$$

because $n > x$ eventually. Thus $(P, d_{\leq})$ is not Yoneda-complete.

For every $\alpha > 0$, the α-Lipschitz inequality for a map $f : P \to \overline{\mathbb{R}}_+$ says exactly that f is order preserving: when $m \leq n$ the right-hand side is zero, and otherwise it is $+\infty$. Since order quasi-metric spaces are standard, α-Lipschitz Yoneda-continuity is equivalent to this inequality together with continuity for the fixed $d_{\leq}$-Scott topology. Both requirements are independent of $\alpha > 0$; see also [2, Proposition 2.5]. Hence

$$\mathcal{L}_\alpha(P, d_{\leq}) = \mathcal{L}_\beta(P, d_{\leq}) = \mathcal{L}_\infty(P, d_{\leq}) \qquad (\alpha, \beta > 0),$$

and the final-topology property follows immediately. □

Remark 7.2. Let X be an infinite set with the discrete metric $d(x, y) = 1$ for $x \neq y$. It is complete, so it has the final-topology property by theorem 1.3. It is not Lipschitz regular: for $2 < 3$, both balls $B(x, < 2)$ and $B(x, < 3)$ equal X, and the former is not relatively compact in the latter.

Thus the necessity picture is:

- for ordinary metric spaces, completeness is necessary and sufficient;
- for continuous quasi-metric spaces, continuity of extension to the formal-ball completion is necessary and sufficient;
- for arbitrary quasi-metric spaces, continuous Yoneda-completeness is a strong sufficient condition but is not logically necessary;
- the quotient-map criterion of theorem 1.1 is exact for all quasi-metric spaces, but a purely internal geometric characterization in the absence of continuity is not asserted here.

8. Consequences and scope

The results can be arranged in three levels.

(1) For every quasi-metric space, the final-topology property is exactly the quotient-map condition of theorem 1.1. This is formally complete but not an intrinsic geometric characterization.

(2) For continuous quasi-metric spaces, the final-topology property is exactly continuity of the simultaneous extension map into the formal-ball completion, equivalently the compact-set condition of theorem 6.4.

(3) For ordinary metric spaces, the final-topology property is exactly metric completeness.

The continuity assumption in the second item is used to construct the way-below map $s(x) = \{b : b \ll (x, 0)\}$ and hence a continuous left inverse to extension. For a completely arbitrary quasi-metric space this map need not be available. Local Yoneda completions provide a possible framework for a further intrinsic characterization [7], but no such theorem is claimed here.

9. Remarks on the hypotheses

The following points record where the hypotheses and the cited results enter the proofs.

(A1) Formal-ball radii are finite elements of $\mathbb{R}_+$, not elements of $\overline{\mathbb{R}}_+$. The extended value $+\infty$ is allowed for distances and function values, but not for radii. All Lipschitz constants used in the directed union are finite and positive.

(A2) The upper-set argument uses that each $\mathcal{L}_n(X, d)$ is a subspace of the Scott space $\mathcal{L}(X)$, so its specialization order is the restricted pointwise order; it does not tacitly assume the pointwise-topology theorem. The later passage to finite-coordinate neighborhoods does use that theorem. The proof writes those conditions explicitly and notes why the lists are nonempty when $\mathbf{0} \notin U$.

(A3) Layer nesting is proved directly from the definitions, so no standardness assumption is imported into the formal characterization for arbitrary quasi-metric spaces. Cone continuity, by contrast, is asserted only under the standardness hypothesis required by the cited cone lemma.

(A4) Finite cone interpolation uses only Scott continuity of $(x, r) \mapsto q(x) - nr$, enough center points, and the minimality of finite maxima of cones. The cone itself is defined by a subtraction-free formula, so the case $d(x, y) = +\infty$ is unambiguous.

(A5) The compactness assertion for the path space is reduced explicitly to a closed subspace of a product of finite discrete spaces in theorem 2.3.

(A6) In the algebraic case, continuity together with Yoneda-completeness makes the formal-ball poset a continuous dcpo, supplying the supremum attached to an infinite branch; standardness, separately, identifies its radius with the infimum of the branch radii and hence with zero.

(A7) Non-Hausdorffness causes no compactness problem: K_k is a continuous image of a compact Hausdorff space. Its specialization saturation Q_k is compact saturated, and lower semicontinuity makes the subbasic condition on K_k equivalent to the one on Q_k.

(A8) The extension from the algebraic case uses only the retract theorem and continuity of precomposition on the relevant function spaces. The retract lemma is stated only for standard spaces, where a Lipschitz Yoneda-continuous map is d-Scott continuous; this hypothesis is satisfied in the main application.

(A9) The completion formula is obtained in formal balls first, then read through the explicit order isomorphism $(z, t) \mapsto (z - t, -t)$. Its image is used only as a sub-dcpo closed under directed suprema in the product; no unnecessary Scott-closedness claim or coordinatewise-supremum assumption is left implicit.

(A10) The value $+\infty$ causes no exceptional case in the cone construction: all cone heights are chosen finite, and the finite root set always admits a uniform positive margin below f.

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